Physics Questions

Isomorphic Multiple-Choice Set

Questions generated below. Multiple-choice answers in RED; incorrect answers in BLUE.

Q1. Which of the following statements is correct?

A. We can model a marathon runner as a point mass when tracking the runner's position over a 10 km race.
This is a valid approximation because the runner's size is negligible compared with the total distance traveled. In large-scale translational motion, we often track only the center of mass, so a point-particle model is appropriate when the object's dimensions are much smaller than the motion scale.

B. We can model a marathon runner as a point mass when analyzing the bending motion of the runner's knee.
This is incorrect because knee motion depends on the runner's detailed geometry and rotation, which a point-mass model cannot represent.

C. We can never imagine the runner moving in a vacuum, even if we want to ignore air resistance in a simplified model.
This confuses an idealized model with literal reality; physicists often neglect drag by using a vacuum approximation when it is useful.

D. None of the above is correct.
This is false because option A gives a standard and appropriate modeling approximation in kinematics.

Answer: *A*

Q2. Which of the following equations is correct?

A. $1\,\text{ms} = 1000\,\text{s}$
This reverses the milli- prefix; milli $= 10^{-3}$, not 10^{3}.

B. $1\,\text{ms} = 0.001\,\text{s}$
This is correct because the prefix milli means 10^{-3}. Therefore $1\,\text{ms} = 10^{-3}\,\text{s} = 0.001\,\text{s}$, and both the number and units are consistent.

C. $1000\,\text{ms} = 0.001\,\text{s}$
This mishandles the conversion factor; in fact $1000\,\text{ms} = 1\,\text{s}$.

D. $1 = 0.001$
This omits units entirely, so it is not a meaningful unit-conversion equation.

Answer: *B*

Q3. Which option has the incorrect unit?

A. Momentum: $\text{kg} \cdot \text{m/s}$
This is a correct SI unit for momentum, so selecting it would confuse a valid derived unit with an invalid one.

B. Pressure: N/m
This is the incorrect unit because pressure is force per area, not force per length. The SI unit of pressure is $\text{Pa} = \text{N/m}^2$, so N/m has the wrong dimensions.

C. Power: watt
This is correct because the SI unit of power is the watt (W).

D. Acceleration: m/s^2
This is the standard SI unit of acceleration, so it is not the incorrect choice.

Answer: *B*

Q4. A rescue drone starts at the origin and detects a beacon at coordinates $\vec{r} = 8\hat{i} + 15\hat{j}$, where distance is measured in kilometers. Which of the following statements is correct?

A. The beacon is 23 km from the origin.
This adds the vector components directly instead of finding the magnitude of the position vector.

B. The beacon is 15 km from the origin.
This uses only the y-component and ignores the x-component in the distance calculation.

C. The beacon is 8 km from the origin.
This uses only the x-component instead of the full two-dimensional displacement.

D. The beacon is 17 km from the origin.
The distance from the origin is the magnitude of the position vector, $|\vec{r}| = \sqrt{8^2 + 15^2}$. Evaluating gives $|\vec{r}| = \sqrt{64 + 225} = \sqrt{289} = 17\,\text{km}$, so this statement is correct.

Answer: *D*

Q5. If the displacement of an object is given by $\vec{s}(t) = [(0.80\,\text{m/s}^3)t^3 + (1.60\,\text{m/s}^2)t^2]\,\hat{i}$, which of the following statements is correct?

A. This is motion with constant acceleration.
A cubic position function produces a time-dependent acceleration, so the acceleration cannot be constant.

B. The magnitude of the acceleration is $1.60\,\text{m/s}^2$.
This mistakes a coefficient in the position function for the actual second derivative needed to find acceleration.

C. The magnitude of the acceleration at $t = 0$ is $3.20\,\text{m/s}^2$.
Differentiate twice: $v_x = \dfrac{ds_x}{dt} = 2.40t^2 + 3.20t$ and $a_x = \dfrac{dv_x}{dt} = 4.80t + 3.20$. At $t = 0$, this gives $a_x(0) = 3.20\,\text{m/s}^2$, so the statement is correct.

D. The magnitude of the acceleration at $t = 0$ is $0\,\text{m/s}^2$.
This comes from differentiating incorrectly or assuming the acceleration must vanish at the start.

Answer: *C*

Q6. An object takes 4.0 s to move from position $\vec{r}_A = 1\hat{i} - 2\hat{j}\,\text{m}$ to position $\vec{r}_B = 9\hat{i} + 4\hat{j}\,\text{m}$ along a straight path. Which of the following statements is correct?

A. The moving distance is 10 m.
Because the path is stated to be straight, the moving distance equals the magnitude of the displacement from A to B. Here $\Delta\vec{r} = (9-1)\hat{i} + (4-(-2))\hat{j} = 8\hat{i} + 6\hat{j}$, so the distance is $|\Delta\vec{r}| = \sqrt{8^2 + 6^2} = 10\,\text{m}$.

B. The speed at $t = 0$ must be $2.5\,\text{m/s}$.
This confuses average speed over the trip with the instantaneous speed at the start, which is not determined from the given data.

C. The acceleration at $t = 0$ must be $0.625\,\text{m/s}^2$.
Distance and travel time alone do not determine the acceleration without an additional model of the motion.

D. None of the above is correct.
This is false because option A follows directly from the straight-line displacement magnitude.

Answer: *A*

Q7. A grasshopper launches itself with speed u at an angle ϕ above the horizontal. Neglecting air resistance, what is the speed of the grasshopper at the maximum height of its trajectory?

A. $u \sin \phi$
This is the initial vertical component of the velocity, and that component becomes zero at the highest point.

B. 0
This assumes the entire velocity vanishes at maximum height, ignoring the nonzero horizontal component.

C. $u \cos \phi$
At maximum height, the vertical component of velocity is zero while the horizontal component remains constant when air resistance is neglected. Since the horizontal component is initially $u \cos \phi$, the speed there is $\sqrt{(u \cos \phi)^2 + 0^2} = u \cos \phi$.

D. u
This ignores the reduction of the vertical component due to gravity during projectile motion.

Answer: *C*

Q8. A drone is initially moving with velocity $\vec{v}_0 = (12.0\,\mathrm{m/s})\hat{i}$. Its motors then produce a constant acceleration $\vec{a} = (1.5\,\mathrm{m/s^2})\hat{i} + (2.5\,\mathrm{m/s^2})\hat{j}$. Which of the following statements is correct?

A. The x-component of the drone's velocity remains constant.
This ignores the nonzero x-acceleration, which changes v_x over time.

B. The y-component of the drone's velocity remains constant.
This ignores the nonzero y-acceleration, which causes v_y to increase from zero.

C. Both the direction of motion and the speed change.
The velocity evolves as $\vec{v}(t) = \vec{v}_0 + \vec{a}t$. Since both $a_x \neq 0$ and $a_y \neq 0$, the velocity components change, which changes the direction; and because the magnitude $|\vec{v}|$ depends on those components, the speed changes as well.

D. None of the above is correct.
This is incorrect because option C correctly describes the effect of a two-dimensional acceleration on the motion.

Answer: *C*

Q9. If the magnitudes of both charges are doubled and the separation between them is halved, what happens to the magnitude of the electrostatic force each exerts on the other?

A. It stays the same.
This ignores that both the product of the charges and the inverse-square distance factor change.

B. It increases by a factor of 4.
This counts only part of the scaling and misses the additional factor from halving the separation.

C. It increases by a factor of 16.
By Coulomb's law, $F = k\frac{|q_1 q_2|}{r^2}$. Doubling both charges multiplies the numerator by $2 \times 2 = 4$, and halving the separation makes the denominator $(r/2)^2 = r^2/4$, which multiplies the force by another factor of 4; overall, the force increases by $4 \times 4 = 16$.

D. It increases by a factor of 8.
This comes from incomplete scaling, typically by handling the charge change correctly but mishandling the squared dependence on distance.

Answer: *C*

Q10. An object can escape a spherical moon of mass M and radius R if it reaches escape speed $v_{\text{esc}} = \sqrt{\dfrac{2GM}{R}}$. Based on this expression, which statement is correct?

A. An object needs a higher escape speed from a moon with greater mass and smaller radius.
The formula $v_{\text{esc}} = \sqrt{\dfrac{2GM}{R}}$ shows that escape speed increases with increasing M and with decreasing R. A larger mass makes the numerator larger and a smaller radius makes the denominator smaller, so both effects raise the required escape speed.

B. A lighter spacecraft needs a lower escape speed from the same moon.
This is incorrect because the spacecraft's own mass does not appear in the escape-speed formula.

C. A moon with smaller mass and larger radius requires a higher escape speed.
This reverses the dependence in the formula, since smaller M and larger R both reduce v_{esc}.

D. Escape speed is independent of the moon's radius.
This ignores the factor of $1/\sqrt{R}$ in the expression for escape speed.

Answer: *A*

Q11. A constant force $\vec{F} = (5.2\,\text{N})\hat{i} - (1.7\,\text{N})\hat{j}$ acts on a particle as it moves along the x-axis from $x = 1.1\,\text{m}$ to $x = 3.6\,\text{m}$. What is the work done by the force on the particle?

A. 17.4 J
This likely uses the magnitude of the full force vector as if both components contributed to work, even though there is no y-displacement.

B. −4.25 J
This incorrectly uses the y-component of force, which does no work because the displacement has no y-component.

C. 13.0 J
Work is the dot product $W = \vec{F} \cdot \Delta\vec{r}$. Since the particle moves only along x, $\Delta x = 3.6 - 1.1 = 2.5\,\text{m}$ and $W = F_x \Delta x = (5.2\,\text{N})(2.5\,\text{m}) = 13.0\,\text{J}$.

D. 3.5 J
This arises from an arithmetic or subtraction error rather than applying $W = F_x \Delta x$ correctly.

Answer: *C*

Q12. A 3.0 kg cart moving at 4.0 m/s strikes a bumper and rebounds at 2.0 m/s in the opposite direction. The collision lasts 0.25 s. What is the magnitude of the average force on the cart during the collision?

A. 24 N
This usually comes from using only one of the speeds instead of the full change in velocity when direction reverses.

B. 48 N
This underestimates the impulse by failing to account correctly for the rebound direction or the full momentum change.

C. 72 N
Use the impulse-momentum theorem: $F_{\text{avg}} = \dfrac{\Delta p}{\Delta t} = \dfrac{m(v_f - v_i)}{\Delta t}$. Taking the initial direction as positive gives $v_i = +4.0\,\text{m/s}$ and $v_f = -2.0\,\text{m/s}$, so $|\Delta v| = 6.0\,\text{m/s}$ and $|F_{\text{avg}}| = \dfrac{(3.0)(6.0)}{0.25} = 72\,\text{N}$.

D. None of the above is correct.
This is false because option C matches the correct impulse calculation.

Answer: *C*

Q13. A 65 kg student standing at rest on a low-friction skateboard catches a 0.50 kg ball thrown horizontally at 20 m/s. What is the student's speed immediately after catching the ball?

A. 0.076 m/s
This is a plausible arithmetic slip from using the wrong total mass or halving the momentum transfer.

B. 0.153 m/s
Because the student catches the ball, this is a perfectly inelastic collision, so linear momentum is conserved: $m_b v_b = (M + m_b) v_f$. Substituting $m_b = 0.50\,\text{kg}$, $v_b = 20\,\text{m/s}$, and $M = 65\,\text{kg}$ gives $v_f = \dfrac{0.50 \times 20}{65.5} \approx 0.153\,\text{m/s}$.

C. 10.0 m/s
This is far too large and ignores that the ball's momentum must be shared with a much more massive person-plus-ball system.

D. 1.31 m/s
This overestimates the final speed, usually by dividing by the student's mass incorrectly or not using the combined mass after the catch.

Answer: *B*
